

Midterm Project – Group2



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Price模型性能:

Metrics	In sample	Out of sample	Cross-validation	Kaggle Score
OLS	0.1184	0.1850	0.21	67.58
Lasso	0.2922	0.3049	0.30	--
Ridge	0.2417	0.2601	0.25	--
ElasticNet	0.2884	0.3006	0.29	--

Rent模型性能:

Metrics	In sample	Out of sample	Cross-validation	Kaggle Score
OLS	0.1401	0.1690	0.18	67.58
Lasso	0.2651	0.2680	0.27	--
Ridge	0.2177	0.2253	0.22	--
ElasticNet	0.2536	0.2577	0.26	--

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房价数据:

模型性能总结 (MAE)

Metrics	In sample	Out of sample	Cross-validation	Kaggle Score
OLS	0.27	0.27	0.27	58.55
LASSO	0.28	0.28	0.28	-
Ridge	0.27	0.27	0.27	-
ElasticNet	0.28	0.28	0.28	-

租金数据:

模型性能总结 (MAE)

Metrics	In sample	Out of sample	Cross-validation	Kaggle Score
OLS	0.25	0.25	0.25	58.55
LASSO	0.25	0.25	0.25	-
Ridge	0.25	0.25	0.25	-
ElasticNet	0.25	0.25	0.25	-

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Model	In-Sample		Out-of-Sample		CV		Kaggle
	Rent	Price	Rent	Price	Rent	Price	
OLS	192,025.92	959,001.37	207,268.86	969,535.37	193,844.39	960,904.38	23.38
Lasso	192,049.24	960,927.58	207,280.57	971,398.49	192,600.09	962,945.91	23.39
Log-OLS	170,645.52	810,639.81	186,588.49	815,630.78	171,285.34	0.3620 (log)	43.91
Random Forest	47,298.21	217,223.72	86,230.96	284,757.10	-	306,092.05	61.22

Merged Model:

Price模型性能:

Metrics	In sample	Out of sample	Cross-validation	Kaggle Score
OLS	0.195048	0.221460	0.22	62.1
Lasso	0.299192	0.311822	0.30	52.9
Ridge	0.253297	0.269940	0.26	55.1
ElasticNet	0.296295	0.308190	0.30	50.4

Rent模型性能:

Metrics	In sample	Out of sample	Cross-validation	Kaggle Score
OLS	0.178869	0.186117	0.19	62.1
Lasso	0.268156	0.269754	0.27	52.9
Ridge	0.229266	0.235477	0.23	55.1
ElasticNet	0.262566	0.266488	0.27	50.4

实事求是